



# **Smart NGO Donation and Volunteer Management Platform**

## **CSA0905 – Programming In Java**

**Guided by:  
Dr. Sivasankar C**

**Presented By:  
G.Deevena(192311391)  
K.Ramya(192311149)**



# Project Overview



## Domain

- NGO Management / Social Impact
- Web-based Donation and Volunteer Management System
- Supports centralized management of NGO activities

## Target Users

- NGO Administrators – manage donors, volunteers, donations, tasks and reports.
- Donors – manage donations and view relevant information.
- Volunteers – view assigned tasks and update task status.

## Project Summary

- A Java + Spring Boot web application for managing donors, donations, volunteers, tasks, and NGO impact.
- Provides separate role-based access for Admin, Donor, and Volunteer.
- Includes Impact Dashboard, notifications, reports, and donation tracking.

## Overall Purpose

- To digitize and centralize NGO operations.
- To reduce manual work and improve coordination.
- To provide transparent, efficient and data-driven NGO management.



# Problem Statement



## Real-World Problem

- NGOs handle large amounts of donor, donation, volunteer, and activity data.
- Many activities are managed using manual records or disconnected systems.
- Tracking the overall social and financial impact can be difficult.

## Current Challenges

- Manual data entry can lead to errors and duplication.
- Difficulty coordinating volunteers and assigning tasks efficiently.
- Limited real-time visibility into donations and NGO activities.
- Generating reports and tracking fund utilization can require significant effort.

## Impact

- Increased administrative workload.
- Reduced operational efficiency and coordination.
- Difficulty maintaining transparency and accountability.
- Delayed decision-making due to lack of centralized information.



# Motivation & Need



## Why the Problem Needs to Be Solved

- NGOs need a centralized digital platform to manage their operations efficiently.
- Automation can reduce paperwork, manual errors, and administrative effort.
- Real-time dashboards and reports can support better decision-making.
- Transparent donation and activity tracking can improve donor trust.

## Industry / Social Relevance

- Digital transformation is increasingly important for non-profit organizations.
- Efficient volunteer coordination helps NGOs respond more effectively to social needs.
- Better donation tracking supports accountability and responsible fund management.
- A centralized platform can help NGOs scale their activities while maintaining operational efficiency.

**“To transform traditional NGO operations into a centralized, transparent, and data-driven digital platform.”**



# Existing System / Current Practice



## Existing Solutions

- NGOs commonly use spreadsheets, paper records, and separate software tools for managing activities.
- Donation and donor information may be maintained in separate records.
- Volunteer coordination is often handled through manual communication and task assignment.

## Existing Approaches

- Manual donor and donation record maintenance.
- Individual tracking of volunteer activities and tasks.
- Periodic report preparation for donations and organizational activities.

## Limitations

- Data duplication and manual errors can occur.
- Lack of a centralized platform connecting donors, volunteers, and administrators.
- Limited real-time monitoring of donations and NGO impact.
- Manual reporting increases administrative effort and time.
- Difficult to provide transparent and up-to-date information to stakeholders.

# Research / Knowledge Gap

## Identified Gap

- Existing approaches do not provide a single integrated platform for donor management, volunteer coordination, and impact monitoring.
- Limited availability of real-time analytics and centralized reporting.
- Donation tracking and volunteer management are often handled as separate processes.
- Automated notifications and task management are not consistently integrated.

## Opportunity for Improvement

- Develop a centralized web platform connecting all major NGO activities.
- Automate routine donation and volunteer management processes.
- Provide real-time impact dashboards and reports.
- Improve transparency through better donation and activity tracking.
- Enable efficient communication through role-based notifications.

**“There is a need for an integrated and data-driven platform that combines NGO donation management, volunteer coordination, and real-time impact monitoring.”**



# Proposed Solution



## Proposed System

Smart NGO Donation and Volunteer Management Platform — a centralized web application developed using Java, Spring Boot, MySQL, Thymeleaf, Bootstrap, and Chart.js.

## Key Features

- **Donor Management:** donor registration, donation recording, donation tracking and receipts.
- **Volunteer Coordination:** volunteer registration, skill management, task assignment and status tracking.
- **Impact Dashboard:** real-time KPIs, donation analytics, category-wise analysis and campaign progress.
- **Notifications & Reports:** role-based notifications and downloadable reports.

## How It Addresses the Gap

- **Centralizes** donor, donation, volunteer and activity information.
- **Automates** repetitive management and coordination tasks.
- Provides **real-time analytics** for better decision-making.
- Improves **transparency, accountability and operational efficiency**.
- Supports different users through **role-based access for Admin, Donor and Volunteer**.



# Objectives



## Primary Objective

- To develop a **centralized web-based platform** for efficient management of NGO donors, donations, volunteers, tasks, and organizational impact.

## Specific & Measurable Objectives

- Develop **role-based access** for Admin, Donor, and Volunteer users.
- Enable administrators to **register and manage donors and volunteers** digitally.
- Provide donation recording, tracking, receipt generation, and **real-time donation analytics**.
- Enable efficient **volunteer task assignment and status tracking**.
- Provide an **Impact Dashboard and reports** for monitoring NGO activities and outcomes.
- Reduce manual record-keeping and improve **transparency, coordination, and operational efficiency**.



# Requirements & Specifications



## Functional Requirements

User registration, login, authentication, and **role-based authorization**.

Donor and donation management with receipt generation.

Volunteer registration, skill management, and task assignment.

Notifications for donations, registrations, and task updates.

Impact Dashboard with KPIs, charts, and campaign progress.

Report generation and CSV export.

## Non-Functional Requirements

- **Security:** BCrypt password encryption and role-based access.
- **Performance:** Efficient database queries and indexing.
- **Usability:** Responsive and user-friendly web interface.
- **Reliability:** Data consistency and error handling.
- **Scalability:** Cloud-deployable architecture supporting multiple users.

## Software Requirements

- **Java 8**
- **Spring Boot 2.7.18 / Spring MVC**
- **Hibernate/JPA**
- **MySQL**
- **Thymeleaf & Bootstrap 5**
- **Chart.js**
- **IntelliJ IDEA / Eclipse, Git & GitHub**

# Literature / Technology



## Research Papers

- Reviewed recent research on **NGO management, donation platforms, volunteer coordination, and digital transformation.**
- Studied approaches for improving **transparency, automation, data management, and stakeholder engagement.**
- Identified the need for integrating multiple NGO functions into a **single platform.**

## Technologies Used

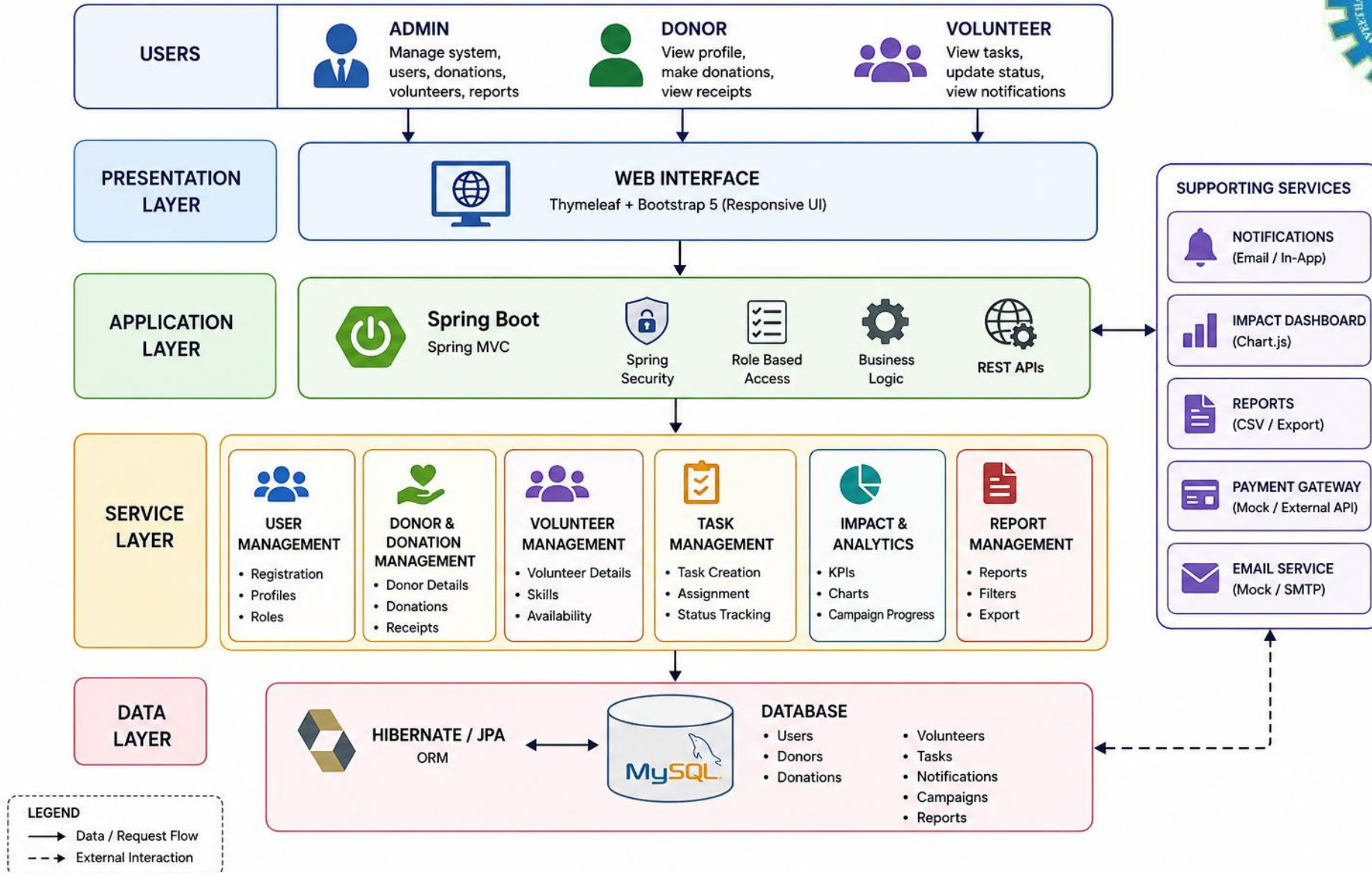
- **Java + Spring Boot:** Backend application development and REST services.
- **Spring MVC + Thymeleaf:** Web application and server-side UI.
- **Hibernate/JPA + MySQL:** Database management and persistence.
- **Bootstrap + Chart.js:** Responsive interface and data visualization.
- **Spring Security + BCrypt:** Authentication and role-based authorization.

## Products / Competing Solutions

- Existing NGO/CRM and donation-management platforms provide individual capabilities such as **fundraising, donor management, or volunteer management.**
- Many solutions are commercial or focus primarily on one operational area.
- The proposed system focuses on **integrating donor management, volunteer coordination, and impact analytics in one platform.**

| Year | Paper Title                                                                                    | Key Contribution                                                                                           | Limitation                                                                    |
|------|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| 2025 | Volunteer Management: A Data Mining Approach                                                   | Proposed a data-driven framework to improve volunteer retention and management using predictive analytics. | Focuses only on volunteer management; does not address donation management.   |
| 2025 | Mobile-Driven Social Impact: An Android Platform for NGO Donation and Volunteer Management     | Developed a mobile application integrating donation and volunteer management.                              | Limited reporting and impact analytics features.                              |
| 2024 | Volunteer Management in Nonprofit Organizations: A Bibliometric Analysis                       | Reviewed global research trends and identified best practices in volunteer management.                     | Literature review only; no implementation of an integrated management system. |
| 2024 | The Effects of Volunteer Management and Personality on Quality of Life and Intention to Donate | Examined how volunteer management influences satisfaction and donation intentions.                         | Focuses on <u>behavioral</u> analysis rather than system development.         |

# SMART NGO PLATFORM – SYSTEM ARCHITECTURE





# Methodology / Workflow



## Development Methodology

- **Agile / Iterative Development** methodology is followed.
- Requirements are identified and divided into individual modules.
- Modules are developed, integrated, tested, and refined incrementally.
- Regular testing and feedback are used to identify and fix issues.

## Project Workflow

**Requirements → System Design → Database Design → UI Development → Backend Implementation → Module Integration → Testing → Deployment**

## Module Workflow

- **Admin:** Manage users → donors → donations → volunteers → tasks → reports → impact analytics.
- **Donor:** Login → manage profile → make/view donations → view receipts → notifications.
- **Volunteer:** Login → view assigned tasks → update task status → notifications.
- **System:** Store data → process activities → generate analytics → provide reports.

# System / Product Design

## Design Components

- **UML Design:** Use Case, Class, and Sequence Diagrams to represent system behavior and interactions.
- **ER Diagram:** Defines relationships between Users, Donors, Volunteers, Donations, Tasks, Campaigns, and Notifications.
- **Database Design:** MySQL database with Hibernate/JPA for data persistence.
- **UI Design:** Responsive web interface with role-based dashboards for Admin, Donor, and Volunteer.
- **Component Architecture:** Presentation, Application, Service, and Data layers.

## Circuit Design

- **Not Applicable** – the project is a software-only web application and does not use electronic hardware.

## Mechanical Design

- **Not Applicable** – no mechanical components are involved.

## Overall Design Flow

Users



Role-Based UI



Spring Boot Application



Service Modules



Hibernate / JPA



MySQL Database



Reports + Analytics + Notifications



# Technology Stack and Tools



## Programming Language

- **Java 8**

## Frameworks

- **Spring Boot 2.7.18**
- **Spring MVC**
- **Spring Security**

## Libraries / UI

- **Hibernate / JPA** – ORM and database persistence
- **Thymeleaf** – server-side web pages
- **Bootstrap 5** – responsive UI
- **Chart.js** – dashboard visualizations
- **BCrypt** – password encryption

## Database

- **MySQL 8**

## APIs

- **Payment API:** Razorpay / Stripe (*or mock payment mode during development*)
- **Notification API:** Email/SMS (*or mock notification mode during development*)

## Development & Deployment Tools

- **IntelliJ IDEA / Eclipse**
- **Git & GitHub**
- **Maven**
- **Docker**
- **Render / Railway / AWS** for cloud deployment



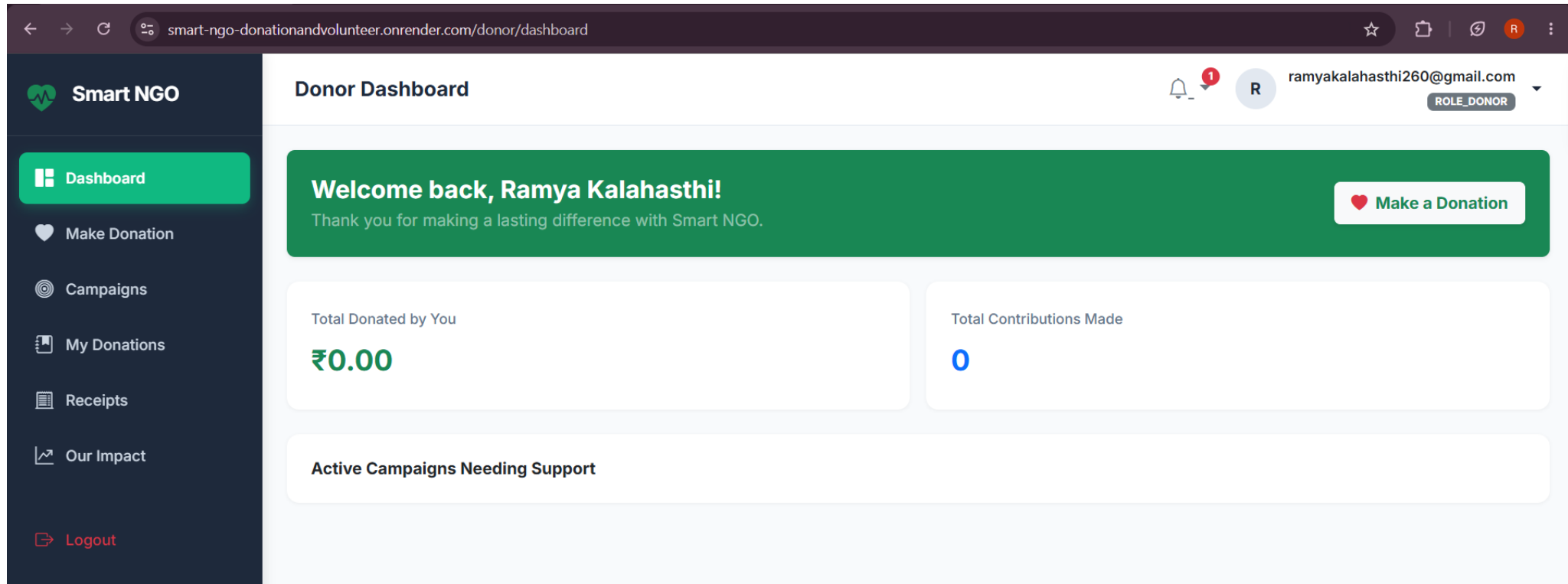
Engineer to Excel

# Module 1 – Donor Management



## Design

- Provides centralized management of donor profiles and donation records.
- Supports both existing donor selection and new donor registration.
- Role-based access ensures secure donor information management.

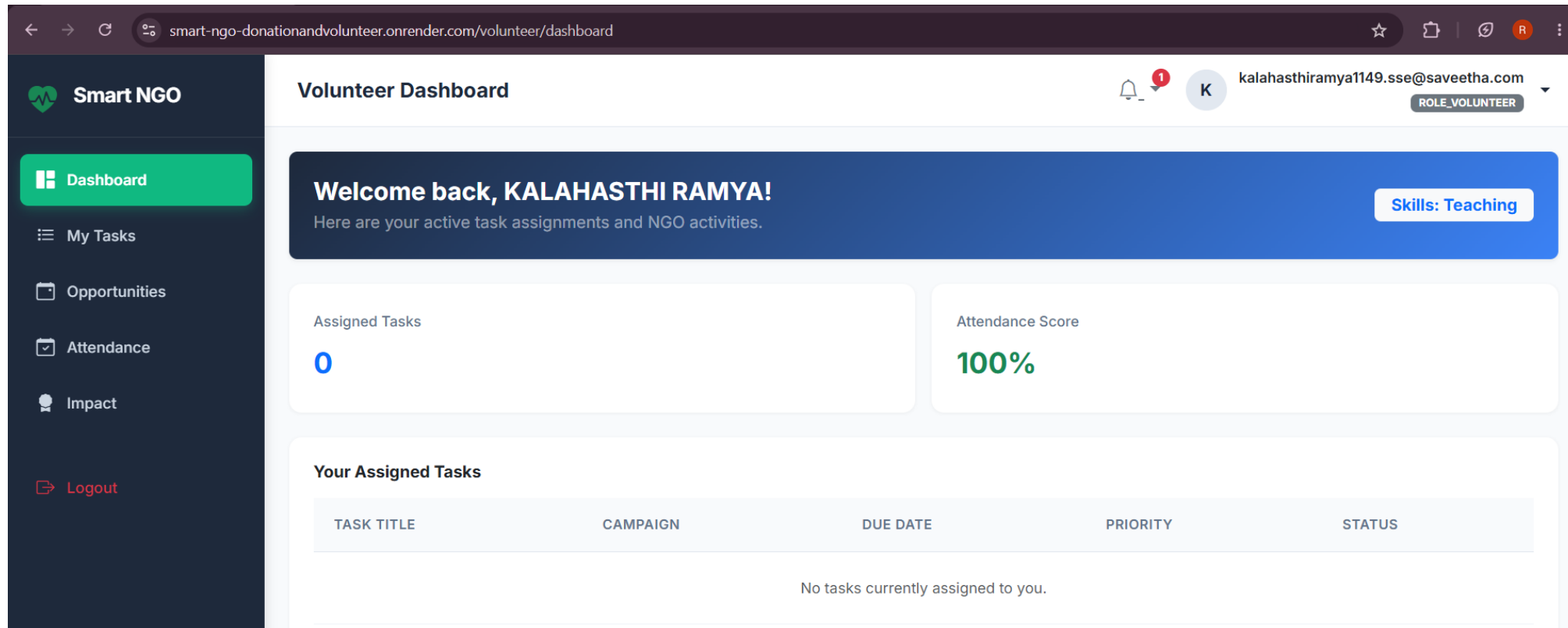


# Module 2 – Volunteer Coordination



## Design

- Maintains volunteer profiles including **skills and contact details**.
- Enables administrators to create volunteers and assign tasks.
- Provides task status tracking: **Assigned** → **In Progress** → **Completed**.



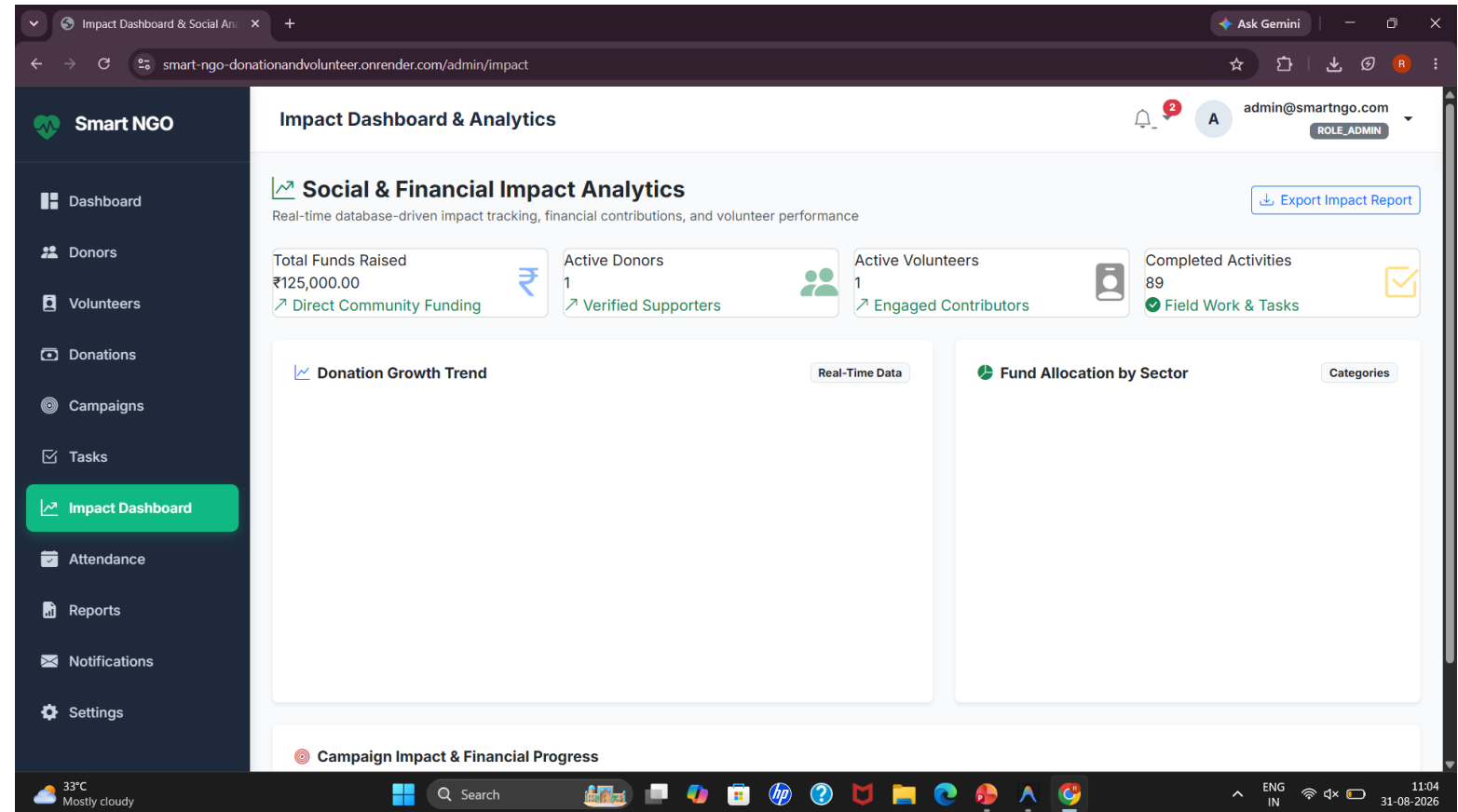
The screenshot shows a web application interface for a volunteer dashboard. The browser address bar displays the URL: `smart-ngo-donationandvolunteer.onrender.com/volunteer/dashboard`. The page has a dark sidebar on the left with the following menu items: "Smart NGO" (with a heart icon), "Dashboard" (highlighted in green), "My Tasks", "Opportunities", "Attendance", "Impact", and "Logout". The main content area is titled "Volunteer Dashboard". At the top right of the main area, there is a notification bell icon with a red "1", a user profile icon with the letter "K", the email address "kalahasthiramy1149.sse@saveetha.com", and a dropdown menu showing "ROLE\_VOLUNTEER". Below this is a blue banner with the text "Welcome back, KALAHASTHI RAMYA!" and "Here are your active task assignments and NGO activities." on the left, and a button labeled "Skills: Teaching" on the right. The dashboard features two summary cards: "Assigned Tasks" showing a large blue "0", and "Attendance Score" showing "100%". Below these is a section titled "Your Assigned Tasks" which contains a table with the following headers: "TASK TITLE", "CAMPAIGN", "DUE DATE", "PRIORITY", and "STATUS". The table body is empty, and a message at the bottom states "No tasks currently assigned to you."

# Module 3 – Impact Dashboard



## Design

- Provides a centralized view of the NGO's **social and financial impact**.
- Displays important KPIs and graphical analytics.
- Designed as a **separate dashboard from the operational Admin Dashboard**.





# Key Algorithm



## Role-Based Access Control

### Donation Processing

Select Existing Donor

OR

Create New Donor



Validate Details



Create Donation Record



Save to MySQL



Generate Receipt



Send Notification



Update Impact Dashboard

## Volunteer Task Workflow

Create Volunteer



Select Skills



Create Task



Assign Volunteer



ASSIGNED



IN\_PROGRESS



COMPLETED

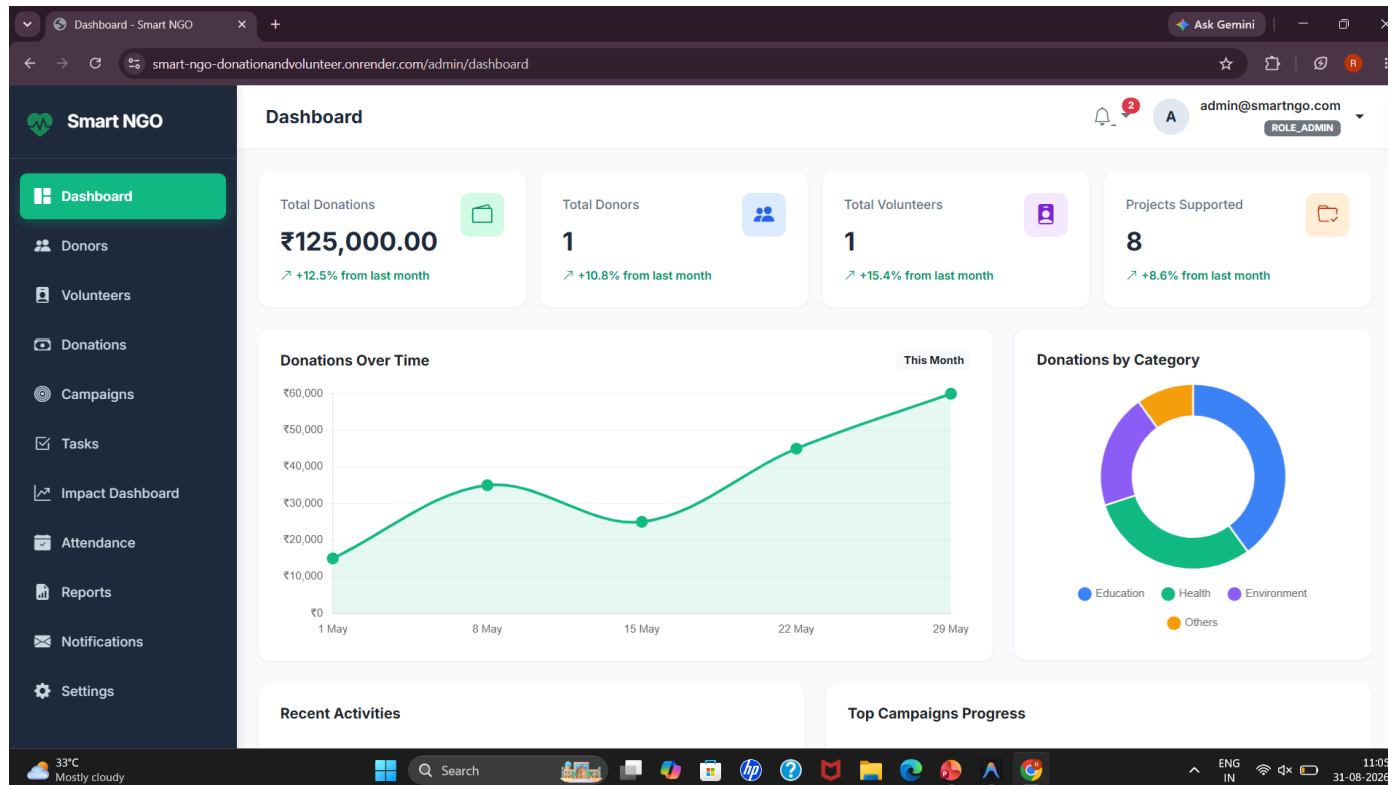


Update Dashboard & Notify Admin

# Prototype / Working Model

## Working Prototype

- Fully functional **web-based NGO management application** developed using Java and Spring Boot.
- Role-based interfaces for **Admin, Donor, and Volunteer**.
- Integrated modules for donor management, volunteer coordination, donations, tasks, notifications, reports, and impact analytics.



# Testing & Validation

## Test Strategy

- Unit testing for individual services and components.
- Integration testing for controller, service, database, and security interactions.
- Manual end-to-end testing for major user workflows.
- Role-based access and authentication validation.

| Test Case          | Expected Result                      | Status   |
|--------------------|--------------------------------------|----------|
| Admin Login        | Admin dashboard displayed            | ✓ Passed |
| Donor Creation     | Donor saved and listed               | ✓ Passed |
| Donation Recording | Donation saved and receipt generated | ✓ Passed |
| Volunteer Creation | Volunteer saved and listed           | ✓ Passed |
| Task Assignment    | Task assigned to volunteer           | ✓ Passed |
| Task Status Update | Status updated successfully          | ✓ Passed |
| Impact Dashboard   | Live metrics/charts displayed        | ✓ Passed |
| Notifications      | Notifications displayed and updated  | ✓ Passed |



# Results & Performance Analysis



## Quantitative Results

- **7/7 automated tests passed — 100% test pass rate.**
- Admin, Donor, and Volunteer dashboard routes verified successfully.
- Core modules successfully integrated with **MySQL**.
- Donation and volunteer records are reflected in the corresponding dashboards.
- Impact Dashboard generates database-driven KPIs and charts.

## Key Metrics

Automated Tests : 7 / 7 Passed  
Core Modules : 3  
User Roles : 3  
Primary Database : MySQL  
Dashboard : Real-time / Database-driven

## Observations

- Centralized data improves accessibility and management.
- Automation reduces repetitive administrative operations.
- Dashboard visualization improves understanding of NGO impact.

# Comparison with Existing Solutions

| Feature              | Existing / Traditional Approach | Proposed System            |
|----------------------|---------------------------------|----------------------------|
| Donor Management     | Manual / Separate Records       | Centralized                |
| Volunteer Management | Manual Coordination             | Digital Task Management    |
| Donation Tracking    | Spreadsheets / Manual           | Integrated Tracking        |
| Impact Monitoring    | Periodic Reports                | Dashboard & Charts         |
| Notifications        | Manual Communication            | System Notifications       |
| Reports              | Manual Preparation              | Automated / CSV Export     |
| Access Control       | Limited                         | Role-Based Access          |
| Data Storage         | Disconnected Records            | Centralized MySQL Database |

## Measurable Improvement

- **7/7 automated tests passed**
- 3 integrated user roles
- 3 major functional modules
- Centralized management of donor, volunteer, donation, and task data
- Automated notification and reporting workflows



# Challenges & Engineering Decisions



## Challenges

- HTTP 500 error during dashboard rendering.
- Notification icon initially lacked functional interaction.
- Impact Dashboard was initially linked to the regular Dashboard.
- New donor creation during donation recording required an additional workflow.
- New volunteer registration encountered validation/database issues.

## Engineering Decisions & Solutions

- Corrected Thymeleaf authentication expression in the shared header.
- Implemented notification dropdown with unread counts and mark-as-read functionality.
- Created a **separate Impact Dashboard** with dedicated analytics.
- Added **inline new donor creation** during donation recording.
- Added transactional volunteer creation with proper validation and error handling.

# Individual Contributions



## Individual Contribution

- Designed the overall **Smart NGO Platform architecture and workflow**.
- Implemented Java + Spring Boot application structure.
- Developed Donor Management, Volunteer Coordination, and Impact Dashboard modules.
- Designed MySQL database entities and Hibernate/JPA integration.
- Implemented Spring Security and role-based access.
- Developed UI using Thymeleaf and Bootstrap.
- Implemented notifications, reports, and dashboard visualizations.
- Performed unit, integration, and end-to-end testing.
- Prepared deployment configuration and documentation.

## Evidence

- Source code / Git commits
- Database design
- UML & ER diagrams
- Application screenshots
- Test results
- Deployment configuration

# Limitations

- Currently dependent on **MySQL** for persistent data storage.
- Payment and notification integrations may operate in **mock mode during development**.
- Public cloud deployment requires configuration of cloud database and hosting credentials.
- Advanced analytics and predictive models are not currently included.
- No dedicated native Android/iOS application; system is currently **web-based**.
- Performance under very large-scale concurrent usage has not yet been benchmarked.



# Future Enhancements



## Scalability

- Deploy using cloud infrastructure for wider accessibility.
- Introduce caching and advanced database optimization for larger datasets.
- Support multiple NGO branches and organizational units.

## Additional Features

- Real-time payment gateway integration.
- Real email/SMS notification services.
- Advanced donor engagement and campaign management.
- Volunteer recommendation based on skills and availability.
- Mobile-responsive Progressive Web App / mobile application.
- Advanced analytics and predictive reporting.



# Conclusion & Project Outcomes



## Objective Achievement

- Successfully developed a centralized NGO management web application.
- Integrated donor management, volunteer coordination, and impact analytics.
- Implemented secure role-based access for Admin, Donor, and Volunteer.
- Automated important operational workflows.



# Project Hosting / Deployment



GitHub Repository: [https://github.com/KalahasthiRamya/Smart\\_NGO\\_DonationAndVolunteer](https://github.com/KalahasthiRamya/Smart_NGO_DonationAndVolunteer)

Live Application: <https://smart-ngo-donationandvolunteer.onrender.com/login>

*Thank  
you!*